

Prashant Rahul

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Work experience

Cypherock, Gurugram (*Remote*)

Aug 2025 – Jan 2026

Software Engineer Intern

- Implemented the Canton Network within STM32 embedded firmware for the X1 crypto hardware wallet, including writing real time encoders and decoders, hashing and signing Canton transactions.
- Contributed to the Electron frontend application, developing features using Typescript.

Open Source & Programs

Linux Kernel Summer 2026 - The Linux Foundation

RTEMS - RTOS for Multiprocessor Systems

- Enabled IMFS testing on BSPs with tight memory constraints, like the sparc/erc32 bsp
- Implemented POSIX compliant ENAMETOOLONG handling in cpukit/libio by validating path components against NAME_MAX during filesystem path evaluation

The Julia Programming Language

- Improved performance in show() for UnitRange append! for AbstractVector
- Quality of life improvements when displaying empty ranges

Education

Bachelor's Degree in Computer Science

2027

CV Raman Global University

Projects

kaworu (*aarch64, kernel, unix-like*)

A WIP unix-like aarch64 kernel

evelin (*compiler, qbe, rust*)

Evelin is a general purpose, statically typed, compiled language using the QBE backend with C FFI support.

cipi8 (*chip8, chip8-emulator, cpp*)

C++ Chip-8 emulator with opcode, graphics, input, and timer support.

zspie (*c, compiler, interpreter*)

Zspie is a fast, easy, dynamic programming language, implemented in C with a virtual machine and opcode based execution.

Skills

Programming languages: Rust, C/C++, Python, Go, LLVM IR, QBE IR, Typescript, Javascript

Technologies & Tools: GNU/Linux, Nix/NixOS, Git, CMake, Make, Meson, Vi/Vim/Neovim, POSIX Shells, NodeJS

Concepts & Practices: Embedded Systems, RTOS, Device drivers, Firmware development, Operating systems, Cryptography Fundamentals, ARM ISA, Version Control, CI/CD